

PAS Tables 2 and 3, recomputed after the estimator and pseudo-ground-truth fixes

$K = 200$ replicates. Real data: default split seed (42; replicate k uses seed $42 + k$), `train_test_split` = 0.2. Synthetic: $m = 200$, $n_j = 20$, $N_j = 80$, split seed 4321, with corrected analytical true second moments (no Monte Carlo moment approximation), $c = 0.05$, and $\psi = 0.2$. Real-data second moments are per problem (`share_var=False`).

Table 1: Post-fix Table 3: results aggregated over $K = 200$ replicates on two real-world datasets with three predictor settings — Amazon review ratings with BERT-base and BERT-tuned predictors, and spiral galaxy fractions with a ResNet50 predictor. Metrics are reported with ± 1 standard error.

Estimator	Amazon (base f)		Amazon (tuned f)		Galaxy	
	MSE ($\times 10^{-3}$)	% Improved $\uparrow$	MSE ($\times 10^{-3}$)	% Improved $\uparrow$	MSE ($\times 10^{-3}$)	% Improved $\uparrow$
Classical	15.607 $\pm$ 0.121	baseline	15.607 $\pm$ 0.121	baseline	1.361 $\pm$ 0.016	baseline
Prediction Avg	41.941 $\pm$ 0.057	23.6 $\pm$ 0.2	4.697 $\pm$ 0.023	69.3 $\pm$ 0.2	7.252 $\pm$ 0.011	13.0 $\pm$ 0.2
PPI	9.973 $\pm$ 0.072	56.8 $\pm$ 0.2	6.262 $\pm$ 0.047	64.4 $\pm$ 0.2	1.024 $\pm$ 0.013	54.0 $\pm$ 0.3
PT	6.679 $\pm$ 0.057	67.6 $\pm$ 0.2	4.893 $\pm$ 0.043	72.9 $\pm$ 0.2	0.681 $\pm$ 0.012	64.9 $\pm$ 0.4
Shrink Classical	10.202 $\pm$ 0.072	54.3 $\pm$ 0.3	2.786 $\pm$ 0.021	77.3 $\pm$ 0.3	0.987 $\pm$ 0.010	50.2 $\pm$ 0.4
Shrink Avg	6.541 $\pm$ 0.059	67.8 $\pm$ 0.2	4.861 $\pm$ 0.044	72.9 $\pm$ 0.2	0.679 $\pm$ 0.012	65.7 $\pm$ 0.3
PAS (ours)	5.603 $\pm$ 0.045	68.7 $\pm$ 0.2	2.949 $\pm$ 0.030	77.5 $\pm$ 0.2	0.617 $\pm$ 0.011	66.1 $\pm$ 0.4
UniPT (ours)	6.214 $\pm$ 0.051	67.2 $\pm$ 0.2	3.678 $\pm$ 0.031	73.3 $\pm$ 0.2	0.622 $\pm$ 0.008	65.3 $\pm$ 0.4
UniPAS (ours)	5.475 $\pm$ 0.044	67.8 $\pm$ 0.2	2.071 $\pm$ 0.016	78.3 $\pm$ 0.2	0.546 $\pm$ 0.006	65.9 $\pm$ 0.3

Table 2: Post-fix Table 2: MSE over $K = 200$ replicates of the synthetic model with the good predictor $f_1(x) = x^2$ and the flawed predictor $f_2(x) = |x|$, using $m = 200$ and the corrected analytical true second moments, $c = 0.05$, and $\psi = 0.2$. UniPT/UniPAS are omitted, as in the paper, since the second moments are known here. Metrics are reported with ± 1 standard error.

Estimator	$f_1(x) = x^2$ MSE ($\times 10^{-3}$)	$f_2(x) = x $ MSE ($\times 10^{-3}$)
Classical	5.187 $\pm$ 0.041	5.187 $\pm$ 0.041
Prediction Avg	2.296 $\pm$ 0.013	38.906 $\pm$ 0.110
PPI	3.402 $\pm$ 0.027	3.492 $\pm$ 0.026
PT	3.172 $\pm$ 0.024	3.090 $\pm$ 0.024
Shrink Classical	1.642 $\pm$ 0.012	4.346 $\pm$ 0.032
Shrink Avg	3.074 $\pm$ 0.023	2.997 $\pm$ 0.022
PAS (ours)	1.623 $\pm$ 0.012	2.898 $\pm$ 0.022

The evaluation target changed; the real-data table is not comparable row-for-row with the published Table 3. The pseudo-ground-truth now averages *all* labels for a problem (Appendix E.4, $\hat{\theta}_j := T_j^{-1} \sum_i \hat{Y}_{ij}$) instead of the unlabelled split only, so every estimator is evaluated against a different pseudo ground-truth. The synthetic target $\theta_j = \eta_j^2$ remains known exactly, but the revised experiment uses $\psi = 0.2$ rather than the published $\psi = 0.1$, so its published table is also included only as a reference.

Second moments. All real-data cells use the per-problem estimators of Appendix C.1, reported as `share_var=False` — the configuration behind the paper’s numbers. On the synthetic model the true second moments are evaluated from their corrected closed forms (Appendix E.1), without a Monte Carlo moment approximation, so that choice does not arise. **Bolding** marks the best value in each column.

Table 3: *Reference only*: the published Table 3, computed before the fixes. Not directly comparable with the post-fix table because the pseudo-ground-truth changed; see the note on the preceding page.

Estimator	Amazon (base f)		Amazon (tuned f)		Galaxy	
	MSE ($\times 10^{-3}$)	% Improved $\uparrow$	MSE ($\times 10^{-3}$)	% Improved $\uparrow$	MSE ($\times 10^{-3}$)	% Improved $\uparrow$
Classical	24.305 $\pm$ 0.189	baseline	24.305 $\pm$ 0.189	baseline	2.073 $\pm$ 0.028	baseline
Prediction Avg	41.332 $\pm$ 0.050	30.7 $\pm$ 0.2	3.945 $\pm$ 0.011	75.4 $\pm$ 0.2	7.195 $\pm$ 0.008	17.0 $\pm$ 0.2
PPI	11.063 $\pm$ 0.085	62.4 $\pm$ 0.2	7.565 $\pm$ 0.066	70.4 $\pm$ 0.2	1.149 $\pm$ 0.017	59.4 $\pm$ 0.3
PT	10.633 $\pm$ 0.089	70.3 $\pm$ 0.2	6.289 $\pm$ 0.050	76.0 $\pm$ 0.2	1.026 $\pm$ 0.015	67.7 $\pm$ 0.3
Shrink Classical	15.995 $\pm$ 0.121	56.4 $\pm$ 0.3	3.828 $\pm$ 0.039	78.9 $\pm$ 0.2	1.522 $\pm$ 0.016	48.8 $\pm$ 0.4
Shrink Avg	9.276 $\pm$ 0.078	70.4 $\pm$ 0.2	6.280 $\pm$ 0.058	77.1 $\pm$ 0.2	0.976 $\pm$ 0.014	68.9 $\pm$ 0.3
PAS (ours)	8.517 $\pm$ 0.071	71.4 $\pm$ 0.2	3.287 $\pm$ 0.024	80.8 $\pm$ 0.2	0.893 $\pm$ 0.011	67.3 $\pm$ 0.4
UniPT (ours)	10.272 $\pm$ 0.084	70.0 $\pm$ 0.2	6.489 $\pm$ 0.053	76.2 $\pm$ 0.2	1.017 $\pm$ 0.015	67.7 $\pm$ 0.3
UniPAS (ours)	8.879 $\pm$ 0.073	69.5 $\pm$ 0.2	3.356 $\pm$ 0.031	77.6 $\pm$ 0.2	0.909 $\pm$ 0.011	66.7 $\pm$ 0.3

Table 4: *Reference only*: the published Table 2, which used $\psi = 0.1$ rather than the revised $\psi = 0.2$.

Estimator	$f_1(x) = x^2$ MSE ($\times 10^{-3}$)	$f_2(x) = x $ MSE ($\times 10^{-3}$)
Classical	3.142 $\pm$ 0.033	3.142 $\pm$ 0.033
Prediction Avg	0.273 $\pm$ 0.004	34.335 $\pm$ 0.147
PPI	2.689 $\pm$ 0.027	2.756 $\pm$ 0.027
PT	2.642 $\pm$ 0.027	2.659 $\pm$ 0.026
Shrink Classical	0.272 $\pm$ 0.003	2.863 $\pm$ 0.030
Shrink Avg	2.486 $\pm$ 0.026	2.537 $\pm$ 0.026
PAS (ours)	0.272 $\pm$ 0.003	2.466 $\pm$ 0.026